Recoverable Ghost States for Bounded-Governed Continual Learning

MeshBrain / MeshModel Research Program

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Abstract

Operational AI systems need memory that can change without allowing hidden state to become authority. Retrieval, long context, adapters, recurrent state, state-space kernels, private commitments, and benchmark ledgers each preserve some form of state, but none is sufficient by itself to decide when remembered or compressed information may govern action. This paper introduces recoverable ghost states: non-authoritative intermediate states that are derived from evidence, queried under policy, projected back into cited packets, and admitted only through verification and governance gates.

The formal model treats state as an estimated and audited object. This follows the state-estimation tradition of partially observable control, the predictive state view that represents systems through action-conditioned predictions of future observations, and provenance models that bind data to entities, activities, agents, derivations, and responsibility. The paper extends those ideas to operational memory: hidden state may inform, but only projected evidence may govern.

The continual-learning contribution is deliberately bounded. We define bounded-governed continual learning as sequential model or memory change that acquires new behavior, preserves protected old behavior, accounts for replay, handles supersession and deletion as valid forgetting, composes declared skills, and emits recoverability certificates with no hard governance regressions. This is not a universal CL claim. It is a manifest-scoped, falsifiable claim form for a declared model family, task stream, horizon, capacity regime, replay budget, retention semantics, governance invariants, and verifier version.

The admitted empirical claims remain bounded to the publication evidence manifest and CL10 final publication admission packet. CL10 admits a bounded-governed CL completion packet; it does not admit universal continual learning, production authority, or transformer replacement. Candidate local artifacts become manuscript evidence only when the claim ledger records the required gate, source artifact, repo commit, command, manifest hash, model hash, dataset hash, seeds, baselines, metric definitions, certificate schema, and failure cases.

Phase45 is admitted as a bounded sidecar result for held-out precursor routing over the same declared local model-family chain. It does not widen CL10 into forecasting, production authority, model-family robustness, or universal CL.

This paper does not claim universal continual learning, serving authority, production authority, endpoint promotion, transformer replacement, or that an external agent-level benchmark result is MeshModel parametric continual-learning proof.

1 Claim Boundary Ledger

State slice: `paper-claim-boundary-ledger`.

The ledger for this manuscript is `docs/paper-claim-ledger.json`. It is the controlling record for paper-claim status. A result category can be admitted only when its required gate passes and the source artifact is bound to

a verified artifact path and hash. CL10 records the bounded completion packet; all non-claim categories remain fixed as non-claims.

The non-claim boundary is fixed:

- Universal continual learning remains not claimed.
- Serving, endpoint promotion, action authority, settlement authority, and production authority remain not claimed.
- Transformer replacement remains not claimed.
- Model-family robustness remains not claimed; CL8 and CL11 are bounded repeated/cross-family evidence only.
- Phase45 proactivity remains a bounded precursor-routing sidecar; it is not a predictive forecasting claim and does not widen CL10.
- CLBench or other external agent-level benchmark evidence remains separate from MeshModel parametric continual-learning proof until admitted by a local intake contract.
- Candidate local evidence may be discussed as a result slot only. It may not be written as a final empirical claim until verifier admission.

2 Introduction

The central problem is not whether a model can remember. The central problem is whether remembered state can be trusted, scoped, invalidated, recovered, and kept from bypassing the authority boundary that owns action.

Operational systems receive logs, policies, incidents, operator notes, evaluation traces, model artifacts, and benchmark results. Those inputs are partial observations of a changing world. A learned state may summarize them, but compression is not the same as evidence. A private commitment may bind a claim, but privacy is not the same as approval. A benchmark score may indicate quality, but quality is not the same as deployment authority. A continual learning update may improve a new task, but improvement is not acceptable if it silently breaks protected behavior, uses unaccounted replay, leaks tenant state, ignores deletion, or promotes stale claims.

Recoverable ghost states address that missing boundary. A ghost state is learned, private, compact, advisory, or intermediate state that is not itself authoritative. It becomes operationally relevant only when it projects into a packet whose sources, hashes, replay accounting, policy scope, and recoverability certificate can be checked.

The governing principle is:

```
Hidden state may inform.
Projected evidence may govern.
Unprojected ghost state may not command.
```

This paper uses MeshBrain for the evidence and authority plane, and MeshModel for the research ghost-kernel plane. MeshBrain records source bundles, policy state, quality gates, promotion ledgers, rollback boundaries, and artifact registries. MeshModel produces benchmark tasks, result packets, replay hashes, controlled CL runs, composition measurements, geometry probes, and publication admission guards.

3 Source-Grounded Positioning

Partially observable decision systems make state estimation unavoidable: decision quality depends on a representation of unobserved state inferred from history. Predictive state representations sharpen this point by grounding state in predictions of future observations conditioned on action sequences rather than in an assumed hidden generative model. Recoverable ghost states adopt the same discipline for operational memory: a state is useful only through the future claims, tests, and transitions it can support.

The W3C provenance model contributes the second discipline. Data is not merely content; it is tied to entities, activities, agents, derivations, attribution, and responsibility. MeshBrain follows that direction with source identifiers, content hashes, bundle hashes, citation references, tenant scope, freshness, and supersession edges. A memory state that cannot be projected back into this provenance surface is not evidence.

Formal verification supplies the third discipline. Separation logic and Iris use ghost state as proof-relevant state that helps reason about programs without becoming ordinary runtime state. This paper imports the separation, not the full logic: learned and private states may be useful to establish an invariant, but they do not mutate authority unless a checked packet enters the authority transition.

Continual reinforcement learning frames learning as ongoing adaptation rather than a one-time solution. The present work narrows that ambition to an operationally testable claim: continual learning is accepted only when the manifest says what should be acquired, what should be retained, what may be forgotten, how replay is accounted, and which governance invariants cannot be violated.

Solomonoff induction supplies a useful boundary, not a result claim. Sterkenburg's 2026 chapter frames universal prediction around computability and argues that the attempted universal-predictor construction does not satisfy the relevant computability desiderata. Its critique of Solomonoff-Levin as practical machine-learning guidance also cautions against treating finite approximations as authority. This paper imports that caution: simplicity and compression can rank plausible worlds, but they cannot promote an unsupported world to evidence or widen bounded CL into universal CL.

External benchmarks remain separate evidence lanes. CLBench is useful because it compares stateful systems with stateless baselines over sequential task instances and reports reward, gain, aggregate reward/gain, and cost. CoIN is useful because it studies continual instruction tuning under task order, instruction diversity, instruction volume, and knowledge preservation. Neither benchmark is automatically MeshModel parametric proof. Each must be admitted through local result schemas before it can support a MeshModel claim.

4 Biomimetic and Mycelium-Inspired ML Precedent

The ghost-state contract is not a biological claim. It does not assert that a model is a fungus, that fungal cognition explains neural network behavior, or that mycelium proves recoverable continual learning. The useful connection is functional: mycelial systems motivate distributed path formation, reinforcement of useful routes, decay or recycling of low-value routes, resilience under local damage, and remediation under environmental constraint. Those mechanisms map naturally onto the engineering vocabulary of sparse subnetworks, adapter paths, recovery certificates, replay accounting, and governed deletion.

Recent mycelium-inspired machine-learning work gives this analogy a concrete prior-art surface. Mycelium-NAS formulates neural architecture search as an emergent growth process, introducing branching, fusion, and orientation as operations for constructing neural architectures while penalizing excessive parameter growth. MycelialNet similarly proposes dynamic connectivity, weak connection pruning, regenerated links, exploration control, and model self-monitoring as mycelial design motifs for data-mining systems. These works are not evidence for MeshModel CL claims, but they show that mycelial growth can be translated into explicit ML mechanisms rather than left as metaphor.

Reservoir-computing work is the closer substrate precedent. Tompris et al. model mycelium growth with cellular automata and reaction-diffusion processes, then evaluate mycelium-inspired reservoir architectures on classification tasks. Related physical-reservoir work treats engineered mycelium morphology as a medium that maps time-varying inputs into nonlinear high-dimensional state trajectories. This aligns with the ghost-state view in one narrow sense: intermediate state can be useful because it preserves a recoverable trajectory over time, not because it is authoritative by itself.

The strongest continual-learning bridge is not explicitly fungal. Sparse CL systems such as NISPA preserve stable paths for older tasks while rewiring plastic paths for new tasks. Grow-only-when-necessary methods expand capacity only when a sequential task stream requires it. These mechanisms supply a direct ML analogue for

a mycelial reading of recoverable ghost states: protected traces should remain stable, new routes should be opened only under a declared gate, and expansion or pruning should be justified by measurable recoverability rather than hidden model convenience.

This paper therefore uses mycelium as a bounded design analogy for four implementation pressures:

- **Branching:** create new adapter or subnetwork paths only when acquisition, retention, or recovery gates fail under the current capacity regime.
- **Fusion:** merge compatible recovered traces into shared paths only when interference tests show that protected behavior remains within tolerance.
- **Reinforcement and decay:** strengthen paths that repeatedly support admitted recovery packets, and prune harmful or inactive paths only after deletion and recoverability policy are satisfied.
- **Damage-tolerant rerouting:** evaluate whether behavior can recover after perturbation, adapter removal, task shift, tenant split, or policy change.

The resulting research stance is biomimetic but claim-bounded: recoverable ghost states are a governed continual-learning analogue of distributed mycelial adaptation, not a biological equivalence claim.

5 Formal Model

Let the system state at time `t` be:

$$S_t = (E_t, G_t, A_t, P_t, L_t, R_t, C_t, U_t, Z_t, B_t, K_t, Pol_t).$$

`E_t` is evidence, `G_t` is ghost state, `A_t` is authoritative operational state, `P_t` is public proof or commitment state, `L_t` is a promotion ledger, `R_t` is a recovery ledger, `C_t` is a conjecture frontier, `U_t` is the admissible test set, `Z_t` is latent predictive state, `B_t` is plausible-world ranking state, `K_t` is the continual-learning stage ledger, and `Pol_t` is policy.

An evidence atom has at least:

$$x = (id, source, content_hash, source_hash, tenant, policy, time, refs).$$

A ghost module writes and reads intermediate state:

$$\gamma_t = W_M(E_{\le t}, c), \qquad y_t = R_M(\gamma_t, q, Pol_t).$$

The invalid transition is:

$$A_{t+1} = T(A_t, \gamma_t).$$

The valid transition consumes only accepted packets:

$$\begin{aligned} Acc_t = \{e \in Pkt : & ProjectOK(e, E_t, G_t, Pol_t) \\ & \wedge VerifyOK(e, E_t, P_t, Pol_t) \\ & \wedge GateOK(e, A_t, Pol_t)\}. \end{aligned}$$

$$A_{t+1} = T_A(A_t, e_t), \qquad e_t \in Acc_t.$$

For continual learning, a stage ledger is:

$$K_i = ((D_1, T_1, m_1), \ldots, (D_i, T_i, m_i)),$$

where `D_i` is a data distribution, `T_i` is a task or behavior family, and `m_i` is the stage manifest. For model state `theta_i`, define acquisition `A_i`, protected retention `R_j`, composition `C_J`, hard regressions `B_i`, replay use `Replay_i`, and certificate `Cert_i`.

The stage predicate is:

$$\begin{aligned} CLOK_i(\theta_i) = A_i(\theta_i) \geq \alpha_i \\ \wedge \forall j < i \text{ protected} : R_j(\theta_i) \geq R_j(\theta_j) - \epsilon_j \\ \wedge Replay_i \leq replay_budget_i \\ \wedge (composition_claimed_i \Rightarrow C_J(\theta_i) \geq \gamma_J) \\ \wedge B_i(\theta_i) = \emptyset \\ \wedge CertificateOK(Cert_i, m_i, R_t, Pol_t). \end{aligned}$$

Old behavior is partitioned before evaluation:

$$Beh_i = Protected_i \cup Superseded_i \cup Deleted_i \cup Unsafe_i \cup Stale_i \cup Obsolete_i \cup Exploratory_i.$$

Valid retention preserves protected behavior, preserves superseded behavior only historically, forbids deleted behavior projection, blocks unsafe behavior, routes stale behavior to revalidation, denies authority to obsolete behavior, and keeps exploratory behavior advisory.

6 Core Theorems

Authority non-bypass. If two executions have the same initial authoritative state and the same accepted-packet sequence, deterministic authority transitions produce the same authority-state sequence regardless of hidden state trajectories.

Evidence conservation. Every accepted packet has non-empty source support for every required claim. This proves citation support, not real-world truth.

Recoverability is stronger than projection. A one-shot packet can project a stale claim. A recoverability certificate requires the claimed invariant to survive perturbation, supersession, policy change, tenant split, replay checks, and projection back into evidence.

Bounded CL is the testable claim. A universal CL statement over all streams, capacities, policies, contradictions, and horizons is not satisfiable by a finite governed learner. A meaningful claim must declare scope, retention semantics, replay budget, capacity regime, governance invariants, and verifier version. This is the claim-boundary counterpart to the Solomonoff lesson: an ideal universal-prediction story does not give a finite governed learner authority outside its declared computable and empirical assumptions.

Certificate soundness. If a complete CL packet is accepted under this contract, acquisition, protected retention, replay-budget compliance, claimed composition, and governed recoverability hold relative to the declared manifest.

7 Local Evidence Ladder

The local experiment ladder is staged so that a later claim cannot silently overwrite an earlier boundary. Every row below is a paper result slot. The ledger decides whether a slot is admitted, blocked, or still a placeholder.

Stage	Local state slice	Status	Evidence boundary
CL0	`cl0-deterministic-recoverability`	placeholder	Deterministic recoverability slot; no final value until admission.
CL1	`cl1-toy-parametric-stream`	placeholder	Toy parametric stream slot; no final value until admission.
CL2	`cl2-adapter-teacher-preflight`	placeholder	Adapter readiness slot; no completed adapter-training claim.
CL3	`cl3-real-small-lm-stream-evidence`	admitted	Single-family small-LM controlled matrix admitted; not model-family robustness or universal CL.
CL4	`phase40-cl4-direct-composition-measurement`	admitted	Direct composition admitted for declared targets only.
CL5	`phase41-cl5-capacity-geometry`	admitted	Adapter-weight geometry admitted; hidden-activation geometry remains unclaimed.
CL6	`phase42-cl6-adversarial-cl`	admitted	Bounded real-model adversarial measurement admitted; general adversarial robustness remains unclaimed.
CL7	`phase43-cl7-paper-result-insertion-guard`	admitted	Publication result insertion guard admitted for bounded evidence only.
CL8	`model-family-repeated-evidence`	admitted	Second-family CL3 repetition admitted; model-family robustness remains not claimed.
CL9	`cl9-external-review-reproduction-admission`	admitted	Reviewer reproduction packet admitted; external benchmark dominance remains not claimed.
CL10	`cl10-final-publication-admission`	admitted	Bounded completion packet admitted; production authority and universal CL remain not claimed.
Phase45	`parametric-cl-proactivity-proof`	admitted sidecar	Held-out precursor routing admitted for the declared local chain; forecasting, production authority, and CL10 widening remain not claimed.
CL11	`cl11-cross-family-cl3-cl6-replication`	admitted	Bounded cross-family CL3-CL6 plus Phase45 replication admitted for three local adapter-ready families; model-family robustness remains unclaimed.

7.1 External Review Evidence Ladder

The live review packet is tracked as a seven-level reviewer ladder in `docs/mesh-live-evidence-ladder-paper-section.md`. L1 through L5 are admitted for public review packaging: local evidence summaries, reviewer reproduction manifest, external review bundle, verify-only reviewer reproduction, and external sidecar summaries. L6 is admitted only as CL12/live-runtime preflight: the CL12 packet verifier passes, but strict live external runtime replication is still blocked on a reachable PrimeIntellect/Slurm/CUDA runtime. L7 is the stable-release verify-only tree; it validates the packaged checkout while recording CL12 live runtime as the sole open gap.

This ladder is review plumbing and claim-boundary evidence. It does not turn CLBench, PrimeIntellect archive material, live-runtime preflight, or stable release packaging into MeshModel parametric continual-learning proof. The paper claim remains controlled by the CL10 admission packet and the claim ledger.

8 Results

8.1 CL3 Controlled Matrix

CL3 admits the single-family small-LM controlled matrix for the declared local model family and task stream. The admitted scope is a repeated seed/order matrix for one local model family, not model-family robustness, production authority, transformer replacement, or universal CL.

Bounded provenance:

```
artifact: docs/meshmodel-phase39-cl3-v5-controlled-matrix-evidence.json
repo_commit: CURRENT_HEAD
command: pnpm exec python3 scripts/run_meshmodel_phase39_cl3_real_small_lm_matrix.py --output-directory a
report_hash: sha256:6c6649bf295452b7e87b2ee36a7b44ba9cd010348a8480366c78c423a17117cb
manifest_hash: sha256:8ba89b374777d08d27d1d4dc4e269248bde0d97fa88cd3fa270ff300559cec7d
model_hash: sha256:f42b8e53d0847afcad7bf51b60585423ef4311fb1c6a2f981e5994998f034a7a
dataset_hash: sha256:9c9452ab09ef3fed5b0c6c766de651c7f195591febcb66bc083959149e1115ef
seeds: [20260512, 20260513, 20260514]
baseline: [naive_sft, sdft_no_replay, adapter_projection, checkpoint_anchor, modular_ghost_state_cl]
metric_definitions: failure_case_count, all_method_metrics_passed, all_single_seed_gates_passed, hard_gov
certificate_schema: mesh.phase39_cl3_real_small_lm_stream_certificate.v1
failure_cases: []
```

Claim boundary: this slot cannot prove model-family robustness, adversarial robustness, publication admission, endpoint promotion, production authority, transformer replacement, or universal CL.

8.2 CL4 Direct Composition

CL4 admits direct one-forward-pass composition evidence for declared targets in the same bounded local model-family chain. It does not claim retrieval-mediated composition, scaffolded composition, production authority, or universal CL.

Bounded provenance:

```
artifact: docs/meshmodel-phase40-cl4-direct-composition-evidence.json
repo_commit: CURRENT_HEAD
command: pnpm exec python3 scripts/run_meshmodel_phase40_cl4_composition_preflight.py --report artifacts/
report_hash: sha256:d19a122e5399df45a6b7b49722d17acd18a1fee9dc6d891b631e51c985105b36
manifest_hash: sha256:eb2b7645e3ee257446cf1c3a0a864895565ea193e295a7ba83d6fe7034e2b829
model_hash: sha256:f42b8e53d0847afcad7bf51b60585423ef4311fb1c6a2f981e5994998f034a7a
dataset_hash: sha256:4ca9c02a2ee6b201d1e42dd7cbc4b2168d29c2ec8b1090322a48fe94ad19d833
seeds: [20260512, 20260513, 20260514]
baseline: [scaffolded_composition_placeholder, retrieval_mediated_composition_placeholder, direct_one_for
metric_definitions: component_task_pass_rate, composition_closure_passed, direct_behavior_score, hard_gov
certificate_schema: mesh.meshmodel_phase40_cl4_direct_composition_certificate.v1
failure_cases: []
```

Claim boundary: this slot cannot generalize beyond its declared target set and cannot establish universal CL, production authority, or transformer replacement.

8.3 CL5 Adapter-Weight Geometry

CL5 admits adapter-weight capacity-geometry evidence tied to the CL4 direct targets. It does not establish hidden-activation geometry, general capacity theory, repeated-family capacity evidence, or model-family robustness.

Bounded provenance:

```
artifact: docs/meshmodel-phase41-cl5-real-geometry-evidence.json
repo_commit: CURRENT_HEAD
command: pnpm exec python3 scripts/run_meshmodel_phase41_cl5_capacity_geometry.py --report artifacts/mesh
report_hash: sha256:2499358bd93c749dbb9760ea42f8269630ee81f854beac073a68748eaeb07f78
manifest_hash: sha256:0fb71fc9826814c1cf2e91c5f4bb699e9e50a5cc211a07f4e31be0f65c52a70f
model_hash: sha256:f42b8e53d0847afcad7bf51b60585423ef4311fb1c6a2f981e5994998f034a7a
dataset_hash: sha256:6dc2ee04cab420e6d9475bcb93c03b787027ff02244ec79a2a2275850e088125
seeds: [20260512, 20260513, 20260514]
baseline: [checkpoint_anchor, modular_ghost_state_cl]
metric_definitions: max_gradient_collision_score, max_projection_norm_removed, expansion_trigger_count, h
certificate_schema: mesh.meshmodel_phase41_cl5_real_geometry_certificate.v1
failure_cases: []
```

Claim boundary: this slot cannot establish hidden-activation geometry, small-LM internal geometry, repeated-family capacity evidence, model-family robustness, production authority, transformer replacement, or universal CL.

8.4 CL6 Bounded Adversarial Measurement

CL6 admits bounded real-model adversarial measurements over declared attack rows. The paper treats this as evidence for the local certificate boundary, not as broad adversarial robustness.

Bounded provenance:

```
artifact: docs/meshmodel-phase42-cl6-real-adversarial-evidence.json
repo_commit: CURRENT_HEAD
command: pnpm exec python3 scripts/run_meshmodel_phase42_cl6_adversarial_cl.py --report artifacts/meshmod
report_hash: sha256:04b6858ad0f55ab13ad1c94b66c1fdd38e4e903f59e94e5e17e98afe7c13f84a
manifest_hash: sha256:86b936556bfeeffce75bbae92f4def50948c9877ed992f8eb0c69f2e1c27e47e
model_hash: sha256:f42b8e53d0847afcad7bf51b60585423ef4311fb1c6a2f981e5994998f034a7a
dataset_hash: sha256:35bf24df2cc1dd732510e196c8a534ca36726a059d3064d64b88cb577a846ebe
seeds: [20260512, 20260513, 20260514]
baseline: [modular_ghost_state_cl]
metric_definitions: measured_attack_run_count, attack_recoverability_pass_rate, hidden_replay_gap_count,
certificate_schema: mesh.meshmodel_phase42_cl6_real_adversarial_certificate.v1
failure_cases: []
```

Claim boundary: this slot cannot establish universal adversarial robustness, serving or production authority, transformer replacement, or universal CL.

8.5 CL8 Model-Family Repetition

CL8 admits a second-family CL3 controlled matrix for Qwen 2.5 0.5B evidence. The admitted statement is repeated model-family evidence, not model-family robustness.

Bounded provenance:

```
artifact: docs/meshmodel-phase44-model-family-repetition-evidence.json
repo_commit: CURRENT_HEAD
command: pnpm exec python3 scripts/run_meshmodel_phase44_model_family_repetition.py --output-directory ar
report_hash: sha256:64ed530b819a73d327e0ca0c681310c88fe57df08657c16a7434e93ad64cca13
manifest_hash: sha256:5879350c915a83e4df26d39759a824126d31ffaa9e12518af062abc3962bffaa
model_hash: sha256:12685f6f00bed036f7270c8167aabb22aadfdb169cb6754ff82c6c34b02ab041
dataset_hash: sha256:9c9452ab09ef3fed5b0c6c766de651c7f195591febcb66bc083959149e1115ef
seeds: [20260512, 20260513, 20260514]
baseline: [primary_single_family_chain, second_family_cl3_controlled_matrix]
```

```
metric_definitions: second_family_executed_run_count, second_family_failure_case_count, hard_governance_r
certificate_schema: mesh.meshmodel_phase44_model_family_repetition_certificate.v1
failure_cases: []
```

Claim boundary: this slot cannot establish model-family robustness, endpoint promotion, production authority, transformer replacement, or universal CL. The supporting repetition claim is the admitted Qwen second-family CL3 controlled matrix; adapter-ready inventory alone is not evidence of robustness.

8.6 CL11 Cross-Family Replication

CL11 admits bounded cross-family replication evidence across three local adapter-ready families under Python 3.14.5. Each admitted family chain has CL3, CL4, CL5, CL6, and Phase45 packets with zero failure cases, zero hard governance regressions, and replay accounting inside budget. The admitted claim is bounded replication across the declared local family set, not model-family robustness.

Bounded provenance:

```
artifact: docs/meshmodel-cl11-cross-family-replication-evidence.json
repo_commit: CURRENT_HEAD
command: PYENV_VERSION=3.14.5 pnpm exec python3 scripts/run_meshmodel_cl11_cross_family_replication.py --
report_hash: sha256:840474298306a6fb5ff5ee8d47ac5601a838e1508db667bfc56bacf992ee2ffd
manifest_hash: sha256:7707ac6ff7e7fcc311de1c8dcc64a47820daadc28ff68f5f6a3b19242404fe47
model_hash: sha256:b6f6b9a907b6162a1c53ebc9e212506f8d01ceb64a45586c115d27f413644eb5
dataset_hash: sha256:b2fc93b9af1b0733c1c98de8cf4ad74c4c2f8145b29b7ac485827313773a639f
seeds: [20260512, 20260513, 20260514]
baseline: [primary_family_chain, second_family_chain, third_family_chain]
metric_definitions: family_count, stage_count_per_family, failure_case_count, hard_governance_regressions
certificate_schema: mesh.cl11_cross_family_replication_certificate.v1
failure_cases: []
```

Claim boundary: this slot establishes bounded cross-family replication evidence only for the three admitted local family chains and their declared CL3-CL6 plus Phase45 packets. Candidate inventory is not proof. Model-family robustness, production authority, transformer replacement, solved continual learning, and universal CL remain not claimed.

8.7 CL10 Final Publication Admission

CL10 admits the bounded completion packet for CL7 through CL9 plus the publication evidence manifest. The packet is a publication admission artifact; it does not grant production readiness, topology causality, external benchmark dominance, model-family robustness, transformer replacement, or universal CL.

Bounded provenance:

```
artifact: docs/meshmodel-cl10-final-publication-admission-evidence.json
packet_path: artifacts/meshmodel-cl10-final-publication-admission-v1/cl10_final_publication_admission_pac
repo_commit: CURRENT_HEAD
command: pnpm exec python3 scripts/verify_cl10_final_publication_admission.py --packet artifacts/meshmode
manifest_hash: sha256:8ba89b374777d08d27d1d4dc4e269248bde0d97fa88cd3fa270ff300559cec7d
manifest_hash: sha256:5879350c915a83e4df26d39759a824126d31ffaa9e12518af062abc3962bffaa
model_hash: sha256:f42b8e53d0847afcad7bf51b60585423ef4311fb1c6a2f981e5994998f034a7a
model_hash: sha256:12685f6f00bed036f7270c8167aabb22aadfdb169cb6754ff82c6c34b02ab041
dataset_hash: sha256:9c9452ab09ef3fed5b0c6c766de651c7f195591febcb66bc083959149e1115ef
seeds: [20260512, 20260513, 20260514]
baseline: publication-evidence-manifest admitted entries
metric_definitions: bounded_governed_cl_completion_packet, blocker_count, upstream_release_decisions
```

```
certificate_schema: mesh.cl10_final_publication_admission_packet.v1
failure_cases: []
```

8.8 Phase45 Bounded Proactivity

Phase45 admits bounded precursor-routing evidence as a sidecar to the local CL3-CL6 chain. The measured statement is narrow: held-out precursor rows are routed before the failure/action point with no future-outcome leakage while protected retention, recoverability, replay accounting, and governance gates remain intact. It is not predictive forecasting, benchmark dominance, production authority, model-family robustness, or a widening of the CL10 completion packet.

Bounded provenance:

```
artifact: docs/meshmodel-phase45-cl-proactivity-evidence.json
repo_commit: CURRENT_HEAD
command: pnpm exec python3 scripts/run_meshmodel_phase45_cl_proactivity.py --output-directory artifacts/m
report_hash: sha256:597a33c2305bbe65e1fafd7c5f3328783b4f501af997eeae8e1db63af4aacc17
manifest_hash: sha256:de1aad185bfee00da541d55a87f9dbab0e08df07286c6e062b7e299f60a47041
model_hash: sha256:f42b8e53d0847afcad7bf51b60585423ef4311fb1c6a2f981e5994998f034a7a
dataset_hash: sha256:dc07325404dbe0cf0bf72138fef85a516452e0c30427cbdda80f2d1547a87344
seeds: [20260512, 20260513, 20260514]
baseline: [reactive_after_failure, modular_ghost_state_cl]
metric_definitions: attack_precursor_row_count, control_precursor_row_count, proactive_acquisition, react
certificate_schema: mesh.meshmodel_phase45_cl_proactivity_certificate.v1
failure_cases: []
```

Claim boundary: this slot cannot establish predictive forecasting, agent-level benchmark dominance, production authority, model-family robustness, transformer replacement, or universal CL.

9 Result Narrative and Visual Evidence

The result is a bounded breakthrough, not an unlimited one. The important change is that continual learning is no longer treated as a slogan about never forgetting. It is treated as a claim that must name the task stream, the protected invariants, the replay budget, the composition floor, the family scope, the verifier, and the failure cases. The experiment succeeds only when the evidence chain says what changed, what stayed protected, what was allowed to be forgotten or superseded, and which gate accepted the result.

The figures below are reading aids for the admitted evidence packets. They do not create new claims. Their source rows are generated from `docs/paper-figures/*.json` and `docs/paper-tables/*.json`, which in turn bind each visual row to report hashes, manifest hashes, source paths, and verifier status.

The main thesis figures are the result-bearing panels: CL3, CL4, CL5, CL6, and the bounded Phase45 sidecar. The evidence-ladder figure also includes the reviewer L4-L7 packaging panel so the paper/PDF shows the live review boundary instead of hiding it in scripts. Administrative and inventory panels remain appendix-style evidence for auditability, not primary empirical support.

Appendix-style audit visuals:

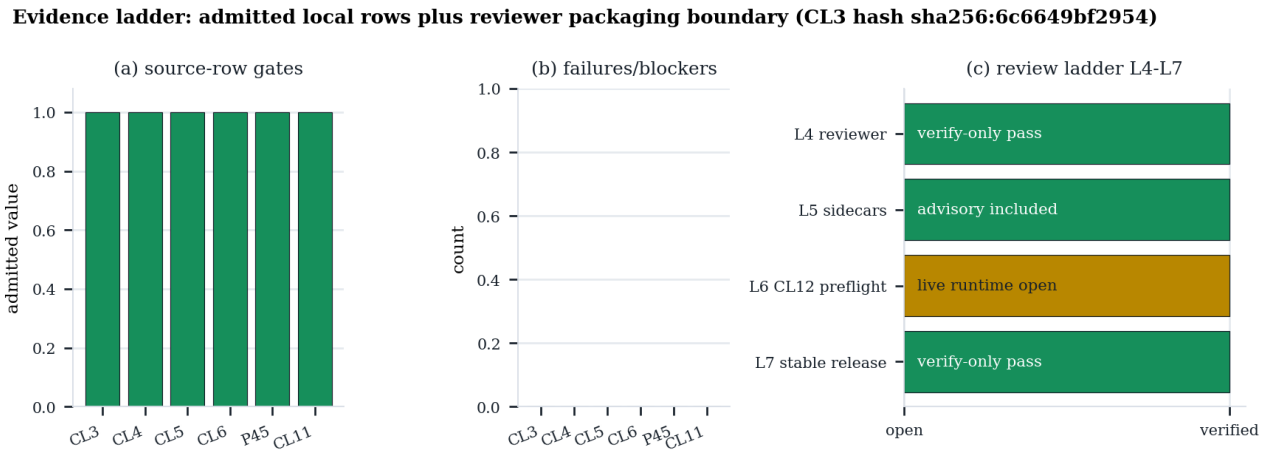


Figure 1: Evidence ladder. The left panels summarize admitted CL3-CL6, Phase45, and CL11 source rows with zero failure/blocker counts; the right panel shows reviewer L4-L7 packaging status and keeps CL12 live runtime as an open preflight gap.

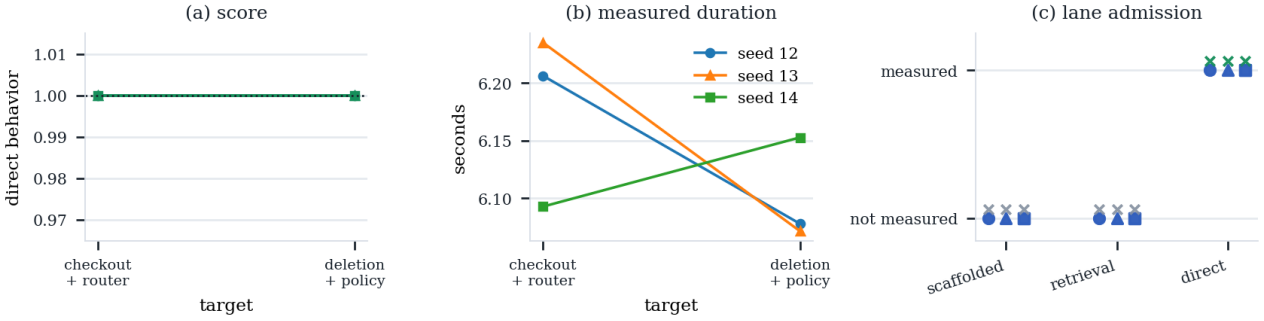


Figure 2: CL4 direct-composition measurement. The panels show direct behavior score, one-forward-pass duration by target and seed, and the non-direct scaffolded/retrieval lanes kept outside the direct-composition claim.

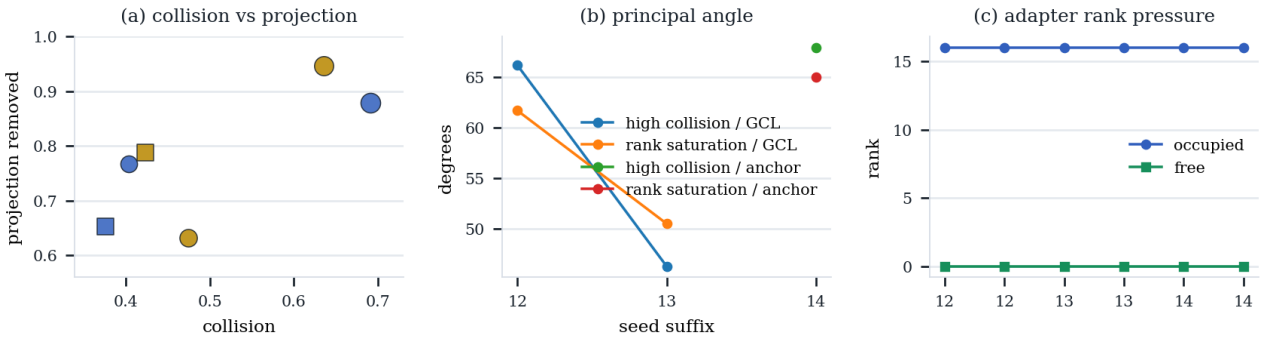


Figure 3: CL5 adapter-weight geometry. The panels show collision versus projection pressure, principal-angle variation by seed and scenario, and adapter-rank pressure; hidden-activation geometry remains outside the admitted claim.

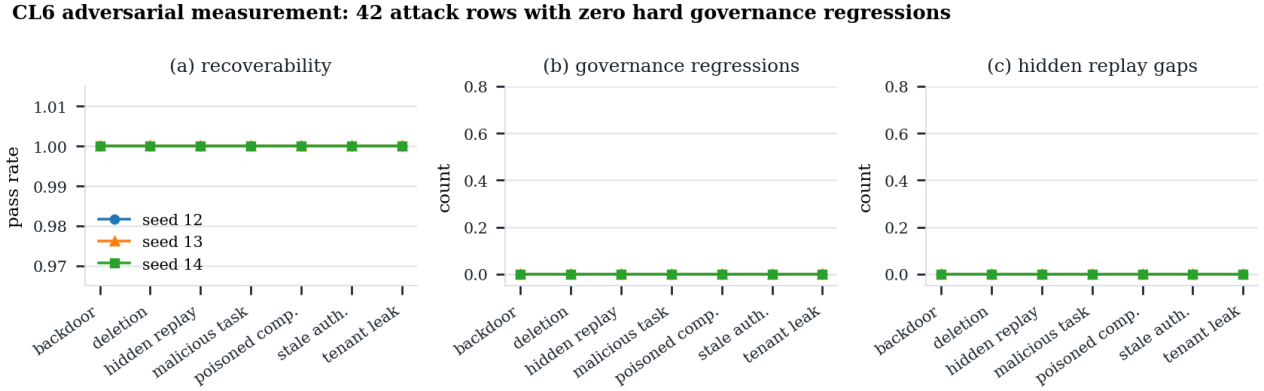


Figure 4: CL6 adversarial measurement. The panels aggregate 42 attack-family rows by seed, showing recoverability at the admitted pass level with zero hard governance regressions and zero hidden replay gaps.

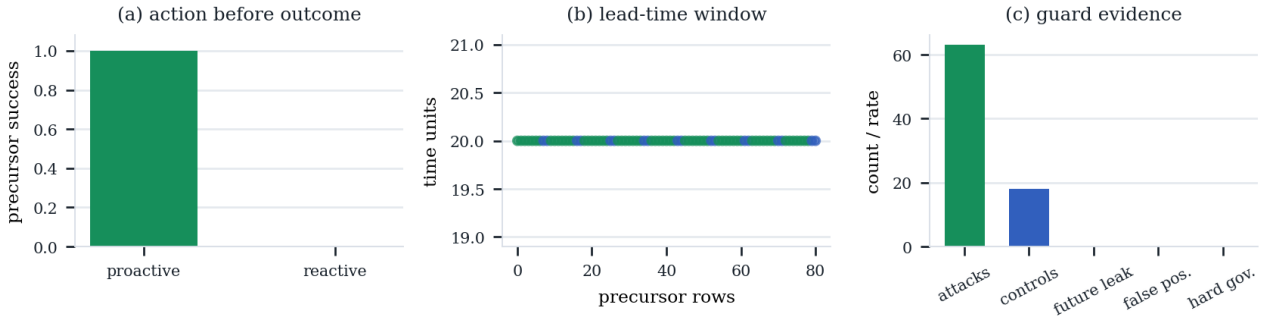


Figure 5: Phase45 bounded proactivity. Held-out precursor routing is compared with the reactive baseline, lead-time windows are shown per precursor row, and future leakage, false positives, and hard governance regressions remain at zero inside the declared sidecar scope.

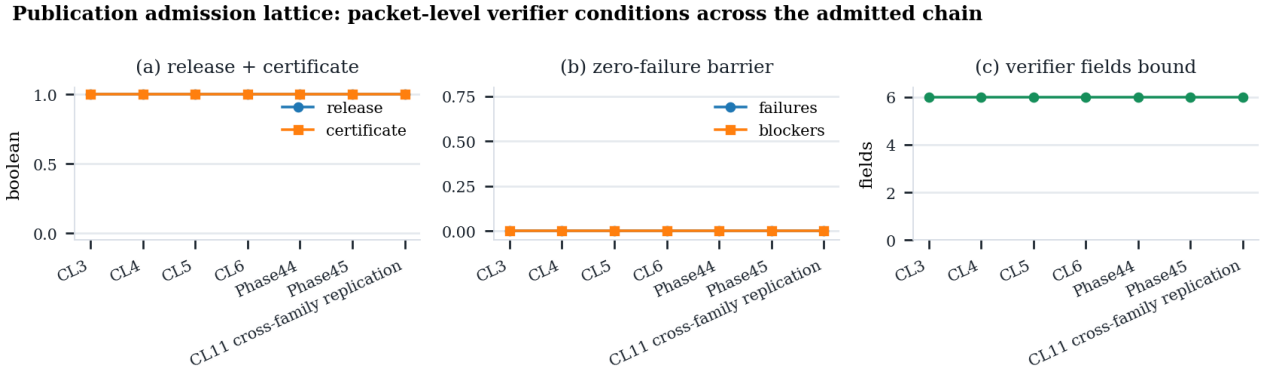


Figure 6: Publication admission lattice. Packet-level verifier conditions remain green across release decision, certificate verification, zero-failure admission, and source-field binding; this is a scoped gate result, not a universal-CL claim.

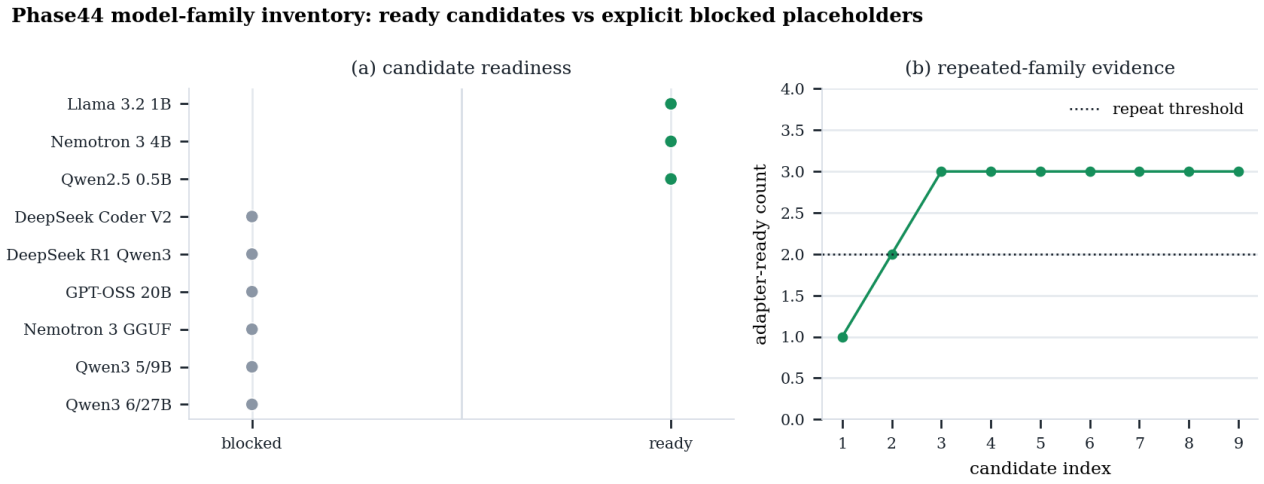


Figure 7: Phase44 model-family inventory and boundaries. Candidate readiness is plotted alongside the cumulative adapter-ready count; the admitted repetition claim comes from the Qwen second-family CL3 controlled matrix, not from inventory alone.

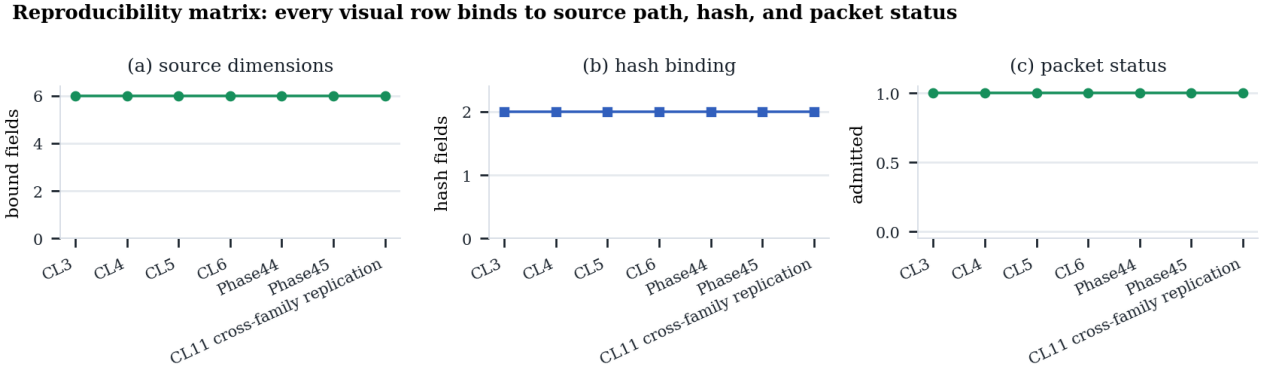


Figure 8: Reproducibility matrix. Each admitted packet is bound to source dimensions, report and manifest hashes, and packet status, preserving the PDF as a reading layer over verifier-bound artifacts.

The incentive for reproducibility is explicit: a reader does not need to trust the prose first. The prose points to the tables and figures; the figures point to source-data rows; the rows point to report and manifest hashes; and the hashes point back to rerunnable verifiers. A result that cannot survive that path remains ghost state. A result that survives it can become an admitted publication claim.

Hypothesis tests remain bounded:

Hypothesis	Current status	Statistical treatment
H1: recoverability separates governed memory from recall	admitted bounded evidence	paired pass-rate interval
H2: CL3 clears acquisition/retention without old raw replay	admitted bounded evidence	bootstrap over seed/order/method cells
H3: declared direct composition can be measured	admitted bounded evidence	binomial interval over target/seed cells
H4: adapter-weight geometry exposes capacity pressure	admitted bounded evidence	correlation and ablation over scenarios
H5: bounded adversarial measurements preserve certificate quality	admitted bounded evidence	attack-row pass accounting inside declared scope
H6: second-family CL3 evidence repeats	admitted bounded evidence	no model-family robustness claim
H7: held-out precursor routing can act before outcomes without leakage	admitted bounded sidecar evidence	precursor-row accounting with reactive baseline and false-positive guard

Sidecar and external-eval rows remain outside the core thesis unless admitted by their own boundary. Phase45 is admitted as bounded parametric proactivity. CLBench local OpenAI-compatible E2E is external agent-level bridge evidence and not MeshModel parametric CL proof. PrimeIntellect archive reanalysis is advisory external archive evidence and not local MeshModel CL proof.

10 Implementation Contract

The local codebase encodes the paper boundary as required manifest fields, result guards, and verifier scripts. The evidence manifest requires:

```
CL_EVIDENCE_REQUIRED_FIELDS = (
    "repo_commit",
    "command",
    "model_hash",
    "model_hash_status",
    "model_family_scope",
    "dataset_hash",
    "seeds",
    "task_order",
    "baseline_ids",
    "metric_definitions",
    "replay_accounting",
    "certificate_schema",
    "failure_cases",
)
```

The publication admission guard requires:

```
PHASE43_CL7_REQUIRED_RESULT_FIELDS = (
    "commit",
    "command",
    "manifest_hash",
    "model_hash",
    "model_family_scope",
    "dataset_hash",
    "seeds",
    "baseline",
    "metric_definitions",
    "certificate_schema",
    "failure_cases",
)
```

This is the central engineering discipline of the paper: a result is not admitted because it exists locally. It is admitted only after its command, commit, hashes, scope, metrics, certificate, and failure cases are visible.

11 Limitations

Current manuscript claims are bounded in six ways.

First, CL3 through CL10 are admitted only through the publication evidence manifest, paper ledger, and CL10 admission packet. Their hashes and commands bind the manuscript to local artifacts; they do not widen the scope beyond declared tasks, model families, metrics, certificates, and failure cases. Solomonoff-style theoretical universality does not widen these admitted artifacts; here it is only an inductive-bias boundary for simplicity and compression in plausible-world ranking.

Second, CL5 measures adapter-weight geometry, not hidden activations. It is useful capacity-pressure evidence, not a full account of the model's internal state geometry.

Third, CL6 is admitted as bounded real-model adversarial measurement. It does not establish general adversarial robustness.

Fourth, CL7 through CL10 admit publication packaging, reviewer reproduction, model-family repetition evidence, and the final bounded completion packet. They do not grant production readiness or endpoint authority.

Fifth, external agent-level evidence remains separate. CLBench now has governed intake and local OpenAI-compatible E2E preflight evidence with corpus isolation, task schedule, provider policy, cost accounting, and mapped metrics, but it remains external agent-level bridge evidence rather than MeshModel parametric model-update proof.

Sixth, topology packets remain sidecar evidence. They do not support topology causality, biological equivalence, or a core CL10 completion requirement.

Seventh, Phase45, CLBench, PrimeIntellect, autonomous speedrun intake, and topology packets are sidecar or advisory unless separately admitted into the publication manifest. They do not widen CL10.

12 Conclusion

Recoverable ghost states give operational memory a disciplined middle layer. The model may learn or compress. The system may rank possible worlds, retain private commitments, or carry adapter deltas. None of that hidden state governs until it projects into evidence and passes verification.

The present manuscript states a bounded admitted result bundle, not a universal continual-learning solution. Bounded-governed CL remains the admissible target form: manifests, source bundles, replay accounting, recovery certificates, composition tests, geometry probes, and publication guards bind before an empirical paragraph can be promoted. The remaining work is clear: broaden family coverage before claiming robustness, measure hidden activations, execute external benchmarks as separate lanes, and keep production authority and transformer replacement out of scope.

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Local Source Anchors

- `docs/ghoststates_drafted.md`
- `docs/meshbrain-meshmodel-ghost-states-paper.md`
- `docs/meshmodel-phase39-cl3-v5-controlled-matrix-evidence.json`
- `docs/meshmodel-phase40-cl4-direct-composition-evidence.json`
- `docs/meshmodel-phase41-cl5-real-geometry-evidence.json`
- `docs/meshmodel-phase44-model-family-repetition-evidence.json`
- `docs/meshmodel-phase45-cl-proactivity-evidence.json`
- `docs/clbench-external-lane.md`
- `mesh_brain/meshmodel/cl_evidence_manifest.py`
- `mesh_brain/meshmodel/cl_real_small_lm_matrix.py`
- `mesh_brain/meshmodel/cl_composition_preflight.py`
- `mesh_brain/meshmodel/cl_capacity_geometry.py`
- `mesh_brain/meshmodel/cl_model_family_repetition.py`
- `mesh_brain/meshmodel/cl_proactivity.py`